DIGITAL LOGIC DESIGN

FULL COMPREHENSIVE QUESTION BANK

CHAPTER 1: INTRODUCTORY CONCEPTS

Short Questions

- What is an Analog System?
- What is a Digital System?
- Define signal and noise.
- Advantages of Digital systems.
- Limitations of Digital systems.

Descriptive Questions

- Compare Analog and Digital systems.
- Discuss advantages of Digital over Analog systems.
- Describe Digital number systems.

CHAPTER 2: NUMBER SYSTEM & CODES

Short Questions

- Define number system.
- Binary, Octal, Decimal, Hexadecimal.
- What is BCD code?
- What is ASCII code?
- Define Byte, Nibble, Word.

Problem-Solving

- Convert Decimal ↔ Binary/Octal/Hex.
- Binary addition and subtraction.
- 2's complement subtraction.

Descriptive Questions

- BCD vs Binary representation.
- Parity method for error detection.
- Design parity generator circuit.

CHAPTER 3: LOGIC GATES & BOOLEAN ALGEBRA

Short Questions

- Define logic gates.
- AND, OR, NOT gates.
- Universal gates.
- Boolean algebra.

Descriptive Questions

- Boolean operations with truth tables.
- Prove De Morgan's Theorem.
- NAND/NOR implementation.
- Alternate gate representation.

Problem-Solving

- Simplify Boolean expressions.
- Apply Boolean laws.

CHAPTER 4: COMBINATIONAL LOGIC

Short Questions

- Combinational circuit.
- Sequential circuit.
- Karnaugh Map (K-map).
- Minterm and Maxterm.

Problem-Solving

- K-map simplification (2–4 variables).
- Grouping (pair, quad, octet).

Descriptive Questions

- SOP vs POS.
- XOR & XNOR circuits.
- Parity generator & checker.

CHAPTER 5: FLIP-FLOPS & SEQUENTIAL LOGIC

Short Questions

- Latch vs Flip-Flop.
- Clock signal.
- Setup & Hold time.

- Race-around condition.

Descriptive Questions

- NAND & NOR latch.
- SR, JK, D, T Flip-Flops.
- Master-Slave Flip-Flop.
- Propagation delay.

CHAPTER 6: ARITHMETIC CIRCUITS

Short Questions

- Binary addition.
- 2’s complement.
- Sign magnitude system.

Problem-Solving

- Binary arithmetic problems.
- 2’s complement operations.

Descriptive Questions

- Half Adder.
- Full Adder.
- Parallel Adder.
- ALU design.

CHAPTER 7: COUNTERS & SHIFT REGISTERS

7.1 COUNTERS - Short Questions

- What is a counter?
- Define propagation delay.
- What is MOD number?

7.1 COUNTERS - Descriptive Questions

- Asynchronous (Ripple) counter.
- Synchronous counter.
- Up/Down counter.
- BCD counter.
- Ring counter.
- Johnson counter.
- 74193 counter.

7.1 COUNTERS - Problem-Solving

- Design MOD-n counters.
- Timing diagrams.
- Cascading counters.

7.2 REGISTERS - Short Questions

- What is a Register?
- What is a Shift Register?
- Serial vs Parallel input.
- SISO, SIPO, PISO, PIPO.
- Universal shift register.

7.2 REGISTERS - Descriptive Questions

- Explain Register with diagram.
- Explain Shift Register operation.
- SISO/SIPO/PISO/PIPO register operation.
- Bidirectional and Universal shift registers.

7.2 REGISTERS - Problem-Solving

- Shifting operation steps.
- Timing diagrams.
- Data storage, transfer, and communication applications.

CHAPTER 8: MSI LOGIC CIRCUITS

Descriptive Questions

- Decoder (3-to-8).
- Encoder (8-to-3).
- BCD-to-Decimal decoder.
- BCD-to-7 segment decoder.

Applications

- Encoder & Decoder uses

CHAPTER 9: MULTIPLEXER & DEMULTIPLEXER

Short Questions

- Multiplexer definition.
- Demultiplexer definition.

Descriptive Questions

- 8:1 Multiplexer.
- Demultiplexer with diagram.
- MUX applications.
- MUX in logic implementation.
- Parallel-to-serial conversion.

CHAPTER 10: ADC & DAC

Short Questions

- ADC definition.
- DAC definition.
- Sampling.

Descriptive Questions

- Digital Ramp ADC.
- Successive Approximation ADC.
- Flash ADC.
- R-2R DAC.
- Sample & Hold circuit.
- Digital Voltmeter.

CHAPTER 11: MEMORY DEVICES

Short Questions

- Memory definition.
- ROM vs RAM.

Descriptive Questions

- ROM architecture.
- EPROM & EEPROM.
- SRAM vs DRAM.
- DRAM operation & refreshing.
- Memory expansion.